

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import math
import missingno as ms
%matplotlib inline
```

```
In [2]: pd.set_option('display.max_rows', 200)
pd.set_option('display.max_columns', 2000)
```

```
In [3]: echonest = pd.read_csv('/Users/khushikhandelwal/fma_metadata/echonest.
features = pd.read_csv('/Users/khushikhandelwal/fma_metadata/features.
genres = pd.read_csv('/Users/khushikhandelwal/fma_metadata/genres.csv'
tracks = pd.read_csv('/Users/khushikhandelwal/fma_metadata/tracks.csv')
```

```
/opt/anaconda3/lib/python3.8/site-packages/IPython/core/interactives
hell.py:3146: DtypeWarning: Columns (0,1,2,3,4,5,6,7,8,11,13,16,17,1
8,19,20,21,22,23,24,25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,
41,42,43,44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,60,61,62,63
,64,65,66,67,68,69,70,71,72,73,74,75,76,77,78,79,80,81,82,83,84,85,8
6,87,88,89,90,91,92,93,94,95,96,97,98,99,100,101,102,103,104,105,106
,107,108,109,110,111,112,113,114,115,116,117,118,119,120,121,122,123
,124,125,126,127,128,129,130,131,132,133,134,135,136,137,138,139,140
,141,142,143,144,145,146,147,148,149,150,151,152,153,154,155,156,157
,158,159,160,161,162,163,164,165,166,167,168,169,170,171,172,173,174
,175,176,177,178,179,180,181,182,183,184,185,186,187,188,189,190,191
,192,193,194,195,196,197,198,199,200,201,202,203,204,205,206,207,208
,209,210,211,212,213,214,215,216,217,218,219,220,221,222,223,224,225
,226,227,228,229,230,231,232,233,234,235,236,237,238,239,240,241,242
,243,244,245,246,247,248,249) have mixed types.Specify dtype option
on import or set low_memory=False.
```

```
has_raised = await self.run_ast_nodes(code_ast.body, cell_name,
/opt/anaconda3/lib/python3.8/site-packages/IPython/core/interactives
hell.py:3146: DtypeWarning: Columns (0,1,2,3,4,5,6,7,8,9,10,11,12,13
,14,15,16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32,33,34,35,3
6,37,38,39,40,41,42,43,44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,
59,60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75,76,77,78,79,80,81
,82,83,84,85,86,87,88,89,90,91,92,93,94,95,96,97,98,99,100,101,102,1
03,104,105,106,107,108,109,110,111,112,113,114,115,116,117,118,119,1
20,121,122,123,124,125,126,127,128,129,130,131,132,133,134,135,136,1
37,138,139,140,141,142,143,144,145,146,147,148,149,150,151,152,153,1
54,155,156,157,158,159,160,161,162,163,164,165,166,167,168,169,170,1
71,172,173,174,175,176,177,178,179,180,181,182,183,184,185,186,187,1
88,189,190,191,192,193,194,195,196,197,198,199,200,201,202,203,204,2
05,206,207,208,209,210,211,212,213,214,215,216,217,218,219,220,221,2
22,223,224,225,226,227,228,229,230,231,232,233,234,235,236,237,238,2
39,240,241,242,243,244,245,246,247,248,249,250,251,252,253,254,255,2
56,257,258,259,260,261,262,263,264,265,266,267,268,269,270,271,272,2
73,274,275,276,277,278,279,280,281,282,283,284,285,286,287,288,289,2
90,291,292,293,294,295,296,297,298,299,300,301,302,303,304,305,306,3
```

```
07,308,309,310,311,312,313,314,315,316,317,318,319,320,321,322,323,3
24,325,326,327,328,329,330,331,332,333,334,335,336,337,338,339,340,3
41,342,343,344,345,346,347,348,349,350,351,352,353,354,355,356,357,3
58,359,360,361,362,363,364,365,366,367,368,369,370,371,372,373,374,3
75,376,377,378,379,380,381,382,383,384,385,386,387,388,389,390,391,3
92,393,394,395,396,397,398,399,400,401,402,403,404,405,406,407,408,4
09,410,411,412,413,414,415,416,417,418,419,420,421,422,423,424,425,4
26,427,428,429,430,431,432,433,434,435,436,437,438,439,440,441,442,4
43,444,445,446,447,448,449,450,451,452,453,454,455,456,457,458,459,4
60,461,462,463,464,465,466,467,468,469,470,471,472,473,474,475,476,4
77,478,479,480,481,482,483,484,485,486,487,488,489,490,491,492,493,4
94,495,496,497,498,499,500,501,502,503,504,505,506,507,508,509,510,5
11,512,513,514,515,516,517,518) have mixed types.Specify dtype optio
n on import or set low_memory=False.
```

```
    has_raised = await self.run_ast_nodes(code_ast.body, cell_name,
/opt/anaconda3/lib/python3.8/site-packages/IPython/core/interactives
hell.py:3146: DtypeWarning: Columns (0,1,5,6,8,12,18,20,21,22,24,33,
34,38,39,44,47,49) have mixed types.Specify dtype option on import o
r set low_memory=False.
```

```
    has_raised = await self.run_ast_nodes(code_ast.body, cell_name,
```

In [4]: `#Analyzing data`

In [5]: `echonest.info()`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 13132 entries, 0 to 13131
Columns: 250 entries, Unnamed: 0 to echonest.248
dtypes: object(250)
memory usage: 25.0+ MB
```

In [6]: `features.info()`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 106577 entries, 0 to 106576
Columns: 519 entries, feature to zcr.6
dtypes: object(519)
memory usage: 422.0+ MB
```

In [7]: `genres.info()`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 163 entries, 0 to 162
Data columns (total 5 columns):
#   Column      Non-Null Count  Dtype
---  -
0   genre_id    163 non-null   int64
1   #tracks     163 non-null   int64
2   parent      163 non-null   int64
3   title       163 non-null   object
4   top_level   163 non-null   int64
dtypes: int64(4), object(1)
memory usage: 6.5+ KB
```

In [8]: `tracks.info()`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 106576 entries, 0 to 106575
Data columns (total 53 columns):
#   Column      Non-Null Count  Dtype
---  -
0   Unnamed: 0  106575 non-null  object
1   album       106575 non-null  object
2   album.1     103046 non-null  object
3   album.2     70295 non-null   object
4   album.3     15296 non-null   object
5   album.4     106575 non-null  object
6   album.5     106575 non-null  object
7   album.6     83150 non-null   object
8   album.7     106575 non-null  object
9   album.8     18061 non-null   object
10  album.9     106575 non-null  object
11  album.10    105550 non-null  object
12  album.11    106575 non-null  object
13  album.12    100067 non-null  object
14  artist      22712 non-null   object
15  artist.1    5376 non-null    object
16  artist.2    14272 non-null   object
17  artist.3    71157 non-null   object
18  artist.4    106575 non-null  object
19  artist.5    105719 non-null  object
20  artist.6    106575 non-null  object
21  artist.7    106575 non-null  object
22  artist.8    44545 non-null   object
23  artist.9    70211 non-null   object
24  artist.10   44545 non-null   object
25  artist.11   46850 non-null   object
26  artist.12   106575 non-null  object
27  artist.13   13153 non-null   object
28  artist.14   106575 non-null  object
29  artist.15   79257 non-null   object
30  artist.16   5582 non-null    object
```

```
31  set          106575 non-null  object
32  set.1        106575 non-null  object
33  track        106575 non-null  object
34  track.1      106575 non-null  object
35  track.2      3671 non-null   object
36  track.3      106575 non-null  object
37  track.4      6160 non-null   object
38  track.5      106575 non-null  object
39  track.6      106575 non-null  object
40  track.7      49599 non-null  object
41  track.8      106575 non-null  object
42  track.9      106575 non-null  object
43  track.10     2350 non-null   object
44  track.11     106575 non-null  object
45  track.12     15025 non-null  object
46  track.13     106488 non-null  object
47  track.14     106575 non-null  object
48  track.15     312 non-null   object
49  track.16     106575 non-null  object
50  track.17     1264 non-null   object
51  track.18     106575 non-null  object
52  track.19     106574 non-null  object
dtypes: object(53)
memory usage: 43.1+ MB
```

```
In [9]: echonest.head(10)
```

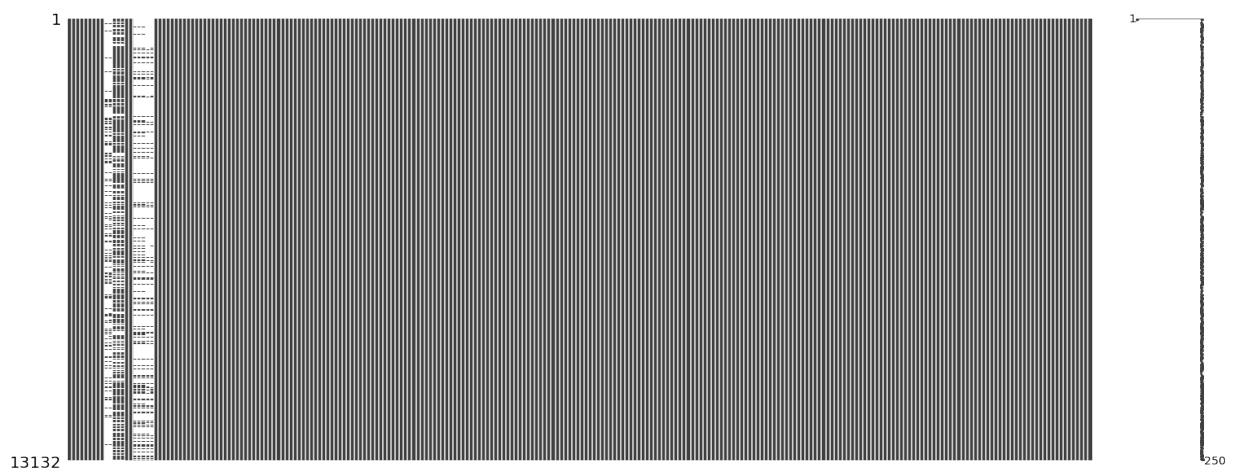
```
Out[9]:
```

	Unnamed: 0	echonest	echonest.1	echonest.2	echonest.3	echonest.4	€
0	NaN	audio_features	audio_features	audio_features	audio_features	audio_features	aud
1	NaN	acousticness	danceability	energy	instrumentalness	liveness	sq
2	track_id	NaN	NaN	NaN	NaN	NaN	
3	2	0.4166752327	0.6758939853	0.6344762684	0.0106280683	0.1776465712	0.1%
4	3	0.3744077685	0.5286430621	0.8174611317	0.0018511032	0.1058799438	0.4%
5	5	0.0435668989	0.7455658702	0.7014699916	0.0006967990	0.3731433124	0.1%
6	10	0.9516699648	0.6581786543	0.9245251615	0.9654270154	0.1154738842	0.0%
7	134	0.4522173071	0.5132380502	0.5604099311	0.0194426943	0.0965666940	0.5%
8	139	0.1065495253	0.2609111726	0.6070668636	0.8350869898	0.2236762711	0.0%
9	140	0.3763124975	0.7340790229	0.2656847734	0.6695811237	0.0859951222	0.0%

```
In [10]: #Feature Engineering
```

```
In [11]: ms.matrix(echonest)
```

```
Out[11]: <AxesSubplot:>
```



```
In [12]: echonest.drop(['echonest.8', 'echonest.9', 'echonest.15', 'echonest.16',
```

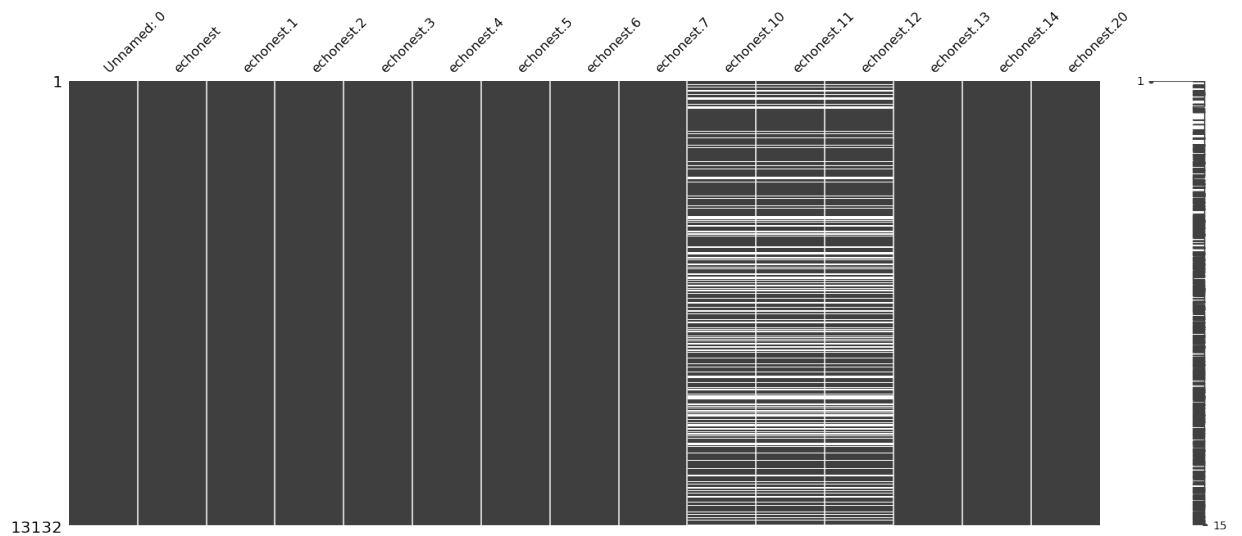
In [13]: `echonest.tail(15)`

Out[13]:

	Unnamed: 0	echonest	echonest.1	echonest.2	echonest.3	echonest.4	echonest.5
13117	124818	0.092864	0.64608	0.772	0.569069	0.442961	0.109521
13118	124819	0.212196	0.691758	0.801736	0.426629	0.231404	0.0584528
13119	124820	0.0595837	0.702435	0.622037	0.099508	0.23637	0.0588595
13120	124850	0.105487	0.690411	0.546009	0.266321	0.205493	0.0316428
13121	124851	0.0348772	0.641127	0.594074	0.478906	0.399061	0.0414732
13122	124852	0.702553	0.7289	0.590636	0.41506	0.118425	0.0373891
13123	124853	0.0203071	0.689575	0.676306	0.891817	0.0959151	0.030968
13124	124854	0.026545	0.667363	0.657399	0.364107	0.108031	0.0428319
13125	124855	0.160281	0.604249	0.554242	0.728696	0.351249	0.175388
13126	124856	0.112268	0.581255	0.665963	0.853041	0.100966	0.0416471
13127	124857	0.00759238	0.790364	0.719288	0.853114	0.720715	0.0825495
13128	124862	0.0414979	0.843077	0.536496	0.865151	0.547949	0.0740006
13129	124863	0.000124355	0.609686	0.895136	0.846624	0.632903	0.0515172
13130	124864	0.327576	0.574426	0.548327	0.452867	0.0759279	0.0333884
13131	124911	0.993606	0.499339	0.0506221	0.945677	0.0959654	0.0651891

```
In [14]: ms.matrix(echonest.iloc[:, 0:15])
```

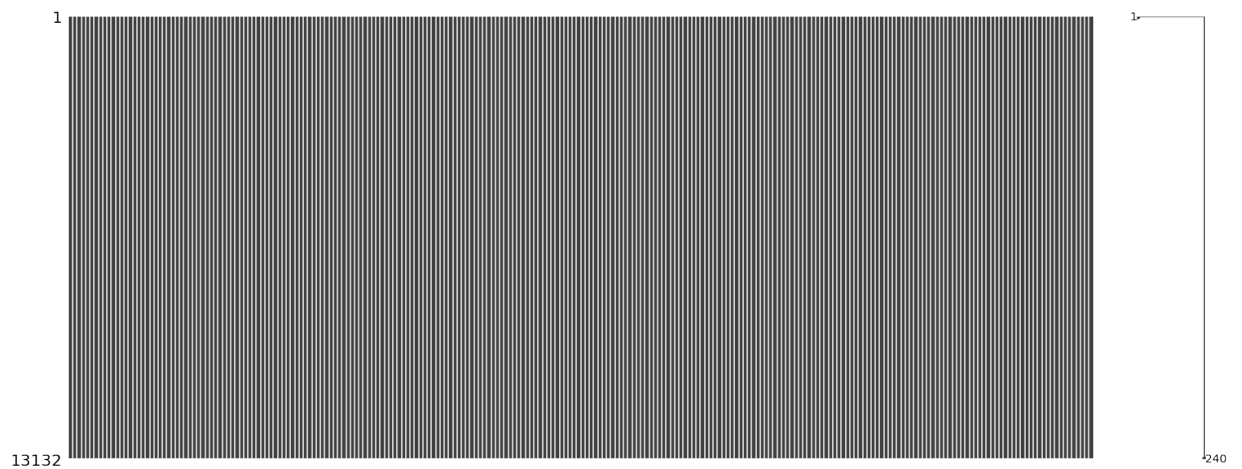
```
Out[14]: <AxesSubplot:>
```



```
In [15]: echonest.drop(['echonest.10', 'echonest.11', 'echonest.12'], axis=1, i
```

```
In [16]: ms.matrix(echonest)
```

```
Out[16]: <AxesSubplot:>
```



```
In [17]: echonest.head(10)
```

```
Out[17]:
```

	Unnamed: 0	echonest	echonest.1	echonest.2	echonest.3	echonest.4	
0	NaN	audio_features	audio_features	audio_features	audio_features	audio_features	auc
1	NaN	acousticness	danceability	energy	instrumentalness	liveness	s
2	track_id	NaN	NaN	NaN	NaN	NaN	
3	2	0.4166752327	0.6758939853	0.6344762684	0.0106280683	0.1776465712	0.1
4	3	0.3744077685	0.5286430621	0.8174611317	0.0018511032	0.1058799438	0.4
5	5	0.0435668989	0.7455658702	0.7014699916	0.0006967990	0.3731433124	0.1
6	10	0.9516699648	0.6581786543	0.9245251615	0.9654270154	0.1154738842	0.0
7	134	0.4522173071	0.5132380502	0.5604099311	0.0194426943	0.0965666940	0.5
8	139	0.1065495253	0.2609111726	0.6070668636	0.8350869898	0.2236762711	0.0
9	140	0.3763124975	0.7340790229	0.2656847734	0.6695811237	0.0859951222	0.0

```
In [18]: echonest.drop(0, axis=0, inplace=True)
```

```
In [19]: echonest.iloc[0, 0]
```

```
Out[19]: nan
```

```
In [20]: echonest.iloc[1, 0]
```

```
Out[20]: 'track_id'
```

```
In [21]: echonest.iloc[0, 0] = echonest.iloc[1, 0]
```


In [22]: `echonest.head()`

Out[22]:

	Unnamed: 0	echonest	echonest.1	echonest.2	echonest.3	echonest.4	ec
1	track_id	acousticness	danceability	energy	instrumentalness	liveness	spe
2	track_id	NaN	NaN	NaN	NaN	NaN	
3	2	0.4166752327	0.6758939853	0.6344762684	0.0106280683	0.1776465712	0.159
4	3	0.3744077685	0.5286430621	0.8174611317	0.0018511032	0.1058799438	0.461
5	5	0.0435668989	0.7455658702	0.7014699916	0.0006967990	0.3731433124	0.124

In [23]: `echonest.drop(2, axis=0, inplace=True)`

In [24]: `echonest.columns = echonest.iloc[0]`

In [25]: `echonest.head()`

Out[25]:

	1	track_id	acousticness	danceability	energy	instrumentalness	liveness	spee
1	track_id	acousticness	danceability	energy	instrumentalness	liveness	spee	
3	2	0.4166752327	0.6758939853	0.6344762684	0.0106280683	0.1776465712	0.159	
4	3	0.3744077685	0.5286430621	0.8174611317	0.0018511032	0.1058799438	0.461	
5	5	0.0435668989	0.7455658702	0.7014699916	0.0006967990	0.3731433124	0.124	
6	10	0.9516699648	0.6581786543	0.9245251615	0.9654270154	0.1154738842	0.032	

In [26]: `echonest.drop(1, axis=0, inplace=True)`

In [27]: `echonest.head()`

Out[27]:

1	track_id	acousticness	danceability	energy	instrumentalness	liveness	speechiness
3	2	0.4166752327	0.6758939853	0.6344762684	0.0106280683	0.1776465712	0.1590000000
4	3	0.3744077685	0.5286430621	0.8174611317	0.0018511032	0.1058799438	0.4610000000
5	5	0.0435668989	0.7455658702	0.7014699916	0.0006967990	0.3731433124	0.1240000000
6	10	0.9516699648	0.6581786543	0.9245251615	0.9654270154	0.1154738842	0.0320000000
7	134	0.4522173071	0.5132380502	0.5604099311	0.0194426943	0.0965666940	0.5250000000

In [28]: `echonest.reset_index(inplace=True)`

In [29]: `echonest.drop('index', inplace=True, axis=1)`

In [30]: `echonest.head()`

Out[30]:

1	track_id	acousticness	danceability	energy	instrumentalness	liveness	speechiness
0	2	0.4166752327	0.6758939853	0.6344762684	0.0106280683	0.1776465712	0.1590000000
1	3	0.3744077685	0.5286430621	0.8174611317	0.0018511032	0.1058799438	0.4610000000
2	5	0.0435668989	0.7455658702	0.7014699916	0.0006967990	0.3731433124	0.1240000000
3	10	0.9516699648	0.6581786543	0.9245251615	0.9654270154	0.1154738842	0.0320000000
4	134	0.4522173071	0.5132380502	0.5604099311	0.0194426943	0.0965666940	0.5250000000

In [31]: `type(echonest['acousticness'][0])`

Out[31]: `str`

```
In [32]: def convert_to_float(df, columns):
          for i in columns:
              df[i] = df[i].astype('float')
          return df
```

```
In [33]: echonest = convert_to_float(echonest, set(echonest.columns) - set(['tr
```

```
In [34]: echonest.head()
```

Out[34]:

	1	track_id	acousticness	danceability	energy	instrumentalness	liveness	speechiness	1
0	2	0.416675	0.675894	0.634476	0.010628	0.177647	0.159310	16	
1	3	0.374408	0.528643	0.817461	0.001851	0.105880	0.461818	12	
2	5	0.043567	0.745566	0.701470	0.000697	0.373143	0.124595	10	
3	10	0.951670	0.658179	0.924525	0.965427	0.115474	0.032985	1	
4	134	0.452217	0.513238	0.560410	0.019443	0.096567	0.525519	1	

```
In [35]: echonest.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 13129 entries, 0 to 13128
Columns: 240 entries, track_id to 223
dtypes: float64(237), object(3)
memory usage: 24.0+ MB
```

```
In [36]: #Working with 'Features' dataset
```

```
In [37]: features.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 106577 entries, 0 to 106576
Columns: 519 entries, feature to zcr.6
dtypes: object(519)
memory usage: 422.0+ MB
```

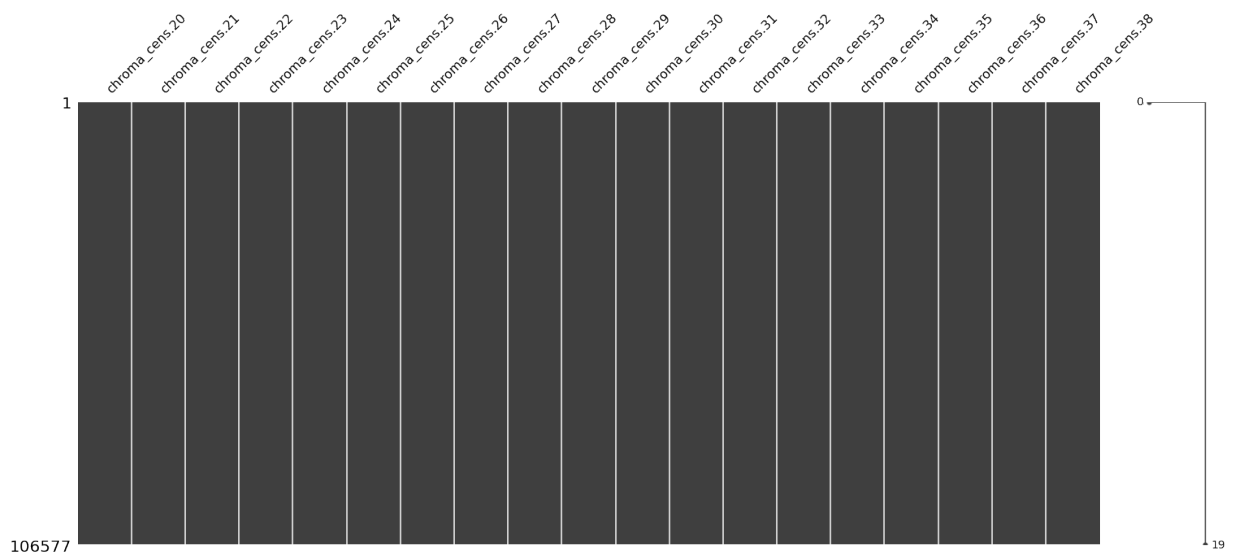
```
In [38]: features.head(10)
```

```
Out[38]:
```

	feature	chroma_cens	chroma_cens.1	chroma_cens.2	chroma_cens.3	ch
0	statistics	kurtosis	kurtosis	kurtosis	kurtosis	
1	number	01	02	03	04	
2	track_id	NaN	NaN	NaN	NaN	
3	2	7.1806526184e+00	5.2303090096e+00	2.4932080507e-01	1.3476201296e+00	1.4824
4	3	1.8889633417e+00	7.6053929329e-01	3.4529656172e-01	2.2952005863e+00	1.6540
5	5	5.2756297588e-01	-7.7654317021e-02	-2.7961030602e-01	6.8588310480e-01	1.9375
6	10	3.7022454739e+00	-2.9119303823e-01	2.1967420578e+00	-2.3444947600e-01	1.3675
7	20	-1.9383698702e-01	-1.9852678478e-01	2.0154602826e-01	2.5855624676e-01	7.752
8	26	-6.9953453541e-01	-6.8415790796e-01	4.8824872822e-02	4.2658798397e-02	-8.1
9	30	-7.2148716450e-01	-8.4855991602e-01	8.9090377092e-01	8.8619679213e-02	-4.4

```
In [39]: ms.matrix(features.iloc[:, 21:40])
```

```
Out[39]: <AxesSubplot:>
```



```
In [40]: #Feature Engineering
```

```
In [41]: features.iloc[0,0] = features.iloc[2, 0]
```

```
In [42]: features.head(3)
```

```
Out[42]:
```

	feature	chroma_cens	chroma_cens.1	chroma_cens.2	chroma_cens.3	chroma_cens.4	ch
0	track_id	kurtosis	kurtosis	kurtosis	kurtosis	kurtosis	
1	number	01	02	03	04	05	
2	track_id	NaN	NaN	NaN	NaN	NaN	

```
In [43]: features.drop(2, inplace=True)
```

```
In [44]: len(features.columns)
```

```
Out[44]: 519
```

```
In [45]: len(features.iloc[0])
```

```
Out[45]: 519
```

```
In [46]: def combine_two_rows(df):
          columns = list(df.columns)
          for i in range(0, 519):
              columns[i] = columns[i] + " " + df.iloc[0, i]
          return columns
```

```
In [47]: features.columns = combine_two_rows(features)
```

```
In [48]: features.drop([0, 1], inplace=True)
```

```
In [49]: features.reset_index(inplace=True)
```

```
In [50]: features.head()
```

```
Out[50]:
```

	index	feature track_id	chroma_cens kurtosis	chroma_cens.1 kurtosis	chroma_cens.2 kurtosis	chroma_cens.3 kurtosis	chroma_cens.4 kurtosis
0	3	2	7.1806526184e+00	5.2303090096e+00	2.4932080507e-01	1.3476201296e+00	
1	4	3	1.8889633417e+00	7.6053929329e-01	3.4529656172e-01	2.2952005863e+00	
2	5	5	5.2756297588e-01	-7.7654317021e-02	-2.7961030602e-01	6.8588310480e-01	
3	6	10	3.7022454739e+00	-2.9119303823e-01	2.1967420578e+00	-2.3444947600e-01	
4	7	20	-1.9383698702e-01	-1.9852678478e-01	2.0154602826e-01	2.5855624676e-01	

```
In [51]: features.drop('index', axis=1, inplace=True)
```

```
In [52]: features.head()
```

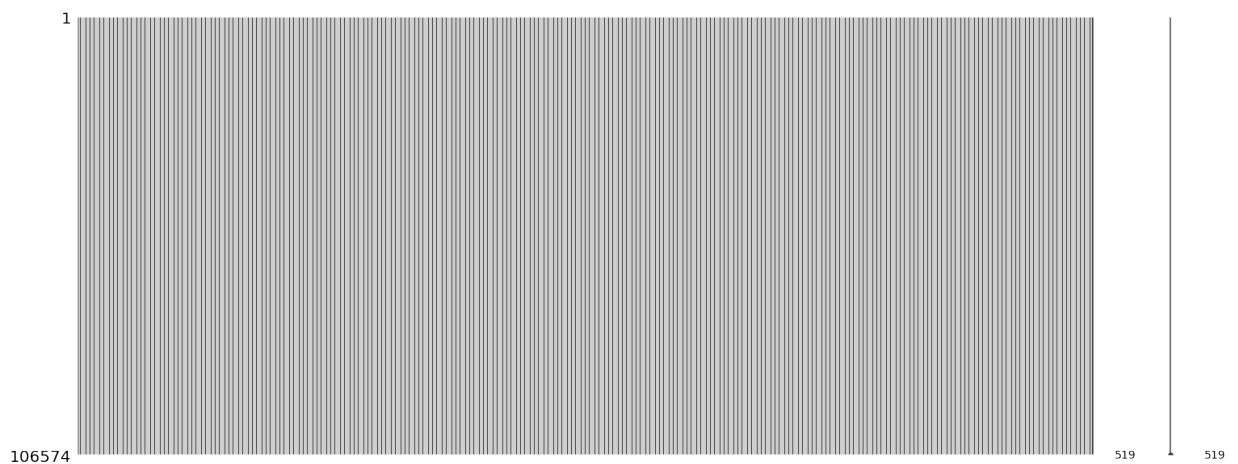
Out[52]:

	feature track_id	chroma_cens kurtosis	chroma_cens.1 kurtosis	chroma_cens.2 kurtosis	chroma_cens.3 kurtosis	chr
0	2	7.1806526184e+00	5.2303090096e+00	2.4932080507e-01	1.3476201296e+00	1.4824
1	3	1.8889633417e+00	7.6053929329e-01	3.4529656172e-01	2.2952005863e+00	1.6540
2	5	5.2756297588e-01	-7.7654317021e-02	-2.7961030602e-01	6.8588310480e-01	1.9375
3	10	3.7022454739e+00	-2.9119303823e-01	2.1967420578e+00	-2.3444947600e-01	1.3673
4	20	-1.9383698702e-01	-1.9852678478e-01	2.0154602826e-01	2.5855624676e-01	7.7521

```
In [53]: features = features.astype(dtype='float')
features['feature track_id'] = features['feature track_id'].astype('int')
```

```
In [54]: ms.matrix(features)
```

Out[54]: <AxesSubplot:>



```
In [55]: features.head(3)
```

```
Out[55]:
```

	feature track_id	chroma_cens kurtosis	chroma_cens.1 kurtosis	chroma_cens.2 kurtosis	chroma_cens.3 kurtosis	chroma_cens.4 kurtosis	ch
0	2	7.180653	5.230309	0.249321	1.347620	1.482478	
1	3	1.888963	0.760539	0.345297	2.295201	1.654031	
2	5	0.527563	-0.077654	-0.279610	0.685883	1.937570	

```
In [56]: #Working with 'Tracks' dataset
```

```
In [57]: tracks.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 106576 entries, 0 to 106575
Data columns (total 53 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Unnamed: 0      106575 non-null object
1   album           106575 non-null object
2   album.1         103046 non-null object
3   album.2         70295 non-null  object
4   album.3         15296 non-null  object
5   album.4         106575 non-null object
6   album.5         106575 non-null object
7   album.6         83150 non-null  object
8   album.7         106575 non-null object
9   album.8         18061 non-null  object
10  album.9         106575 non-null object
11  album.10        105550 non-null object
12  album.11        106575 non-null object
13  album.12        100067 non-null object
14  artist          22712 non-null  object
15  artist.1        5376 non-null   object
16  artist.2        14272 non-null  object
17  artist.3        71157 non-null  object
18  artist.4        106575 non-null object
19  artist.5        105719 non-null object
20  artist.6        106575 non-null object
21  artist.7        106575 non-null object
22  artist.8        44545 non-null  object
23  artist.9        70211 non-null  object
24  artist.10       44545 non-null  object
25  artist.11       46850 non-null  object
26  artist.12       106575 non-null object
27  artist.13       13153 non-null  object
28  artist.14       106575 non-null object
29  artist.15       79257 non-null  object
```

```
30  artist.16      5582 non-null    object
31  set            106575 non-null  object
32  set.1          106575 non-null  object
33  track          106575 non-null  object
34  track.1        106575 non-null  object
35  track.2        3671 non-null    object
36  track.3        106575 non-null  object
37  track.4        6160 non-null    object
38  track.5        106575 non-null  object
39  track.6        106575 non-null  object
40  track.7        49599 non-null   object
41  track.8        106575 non-null  object
42  track.9        106575 non-null  object
43  track.10       2350 non-null    object
44  track.11       106575 non-null  object
45  track.12       15025 non-null   object
46  track.13       106488 non-null  object
47  track.14       106575 non-null  object
48  track.15       312 non-null     object
49  track.16       106575 non-null  object
50  track.17       1264 non-null    object
51  track.18       106575 non-null  object
52  track.19       106574 non-null  object
dtypes: object(53)
memory usage: 43.1+ MB
```


In [58]: `tracks.head()`

Out[58]:

	Unnamed: 0	album	album.1	album.2	album.3	album.4	album.5	album.6
0	NaN	comments	date_created	date_released	engineer	favorites	id	information
1	track_id	NaN	NaN	NaN	NaN	NaN	NaN	NaN
2	2	0	2008-11-26 01:44:45	2009-01-05 00:00:00	NaN	4	1	
3	3	0	2008-11-26 01:44:45	2009-01-05 00:00:00	NaN	4	1	
4	5	0	2008-11-26 01:44:45	2009-01-05 00:00:00	NaN	4	1	

In [59]: `tracks.iloc[0,0] = tracks.iloc[1, 0]`

In [60]: `tracks.drop(1, axis=0, inplace=True)`

```
In [61]: tracks.head()
```

```
Out[61]:
```

	Unnamed: 0	album	album.1	album.2	album.3	album.4	album.5	album.6
0	track_id	comments	date_created	date_released	engineer	favorites	id	information
2	2	0	2008-11-26 01:44:45	2009-01-05 00:00:00	NaN	4	1	
3	3	0	2008-11-26 01:44:45	2009-01-05 00:00:00	NaN	4	1	
4	5	0	2008-11-26 01:44:45	2009-01-05 00:00:00	NaN	4	1	
5	10	0	2008-11-26 01:45:08	2008-02-06 00:00:00	NaN	4	6	NaN

```
In [62]: #Feature Engineering
```

```
In [63]: len(tracks.columns)
```

```
Out[63]: 53
```

```
In [64]: def combine_one_row(df):
          columns = list(df.columns)
          for i in range(0, 53):
              if i == 0:
                  columns[i] = df.iloc[0, i]
              else:
                  columns[i] = columns[i] + " " + df.iloc[0, i]
          return columns
```

```
In [65]: tracks.columns = combine_one_row(tracks)
```

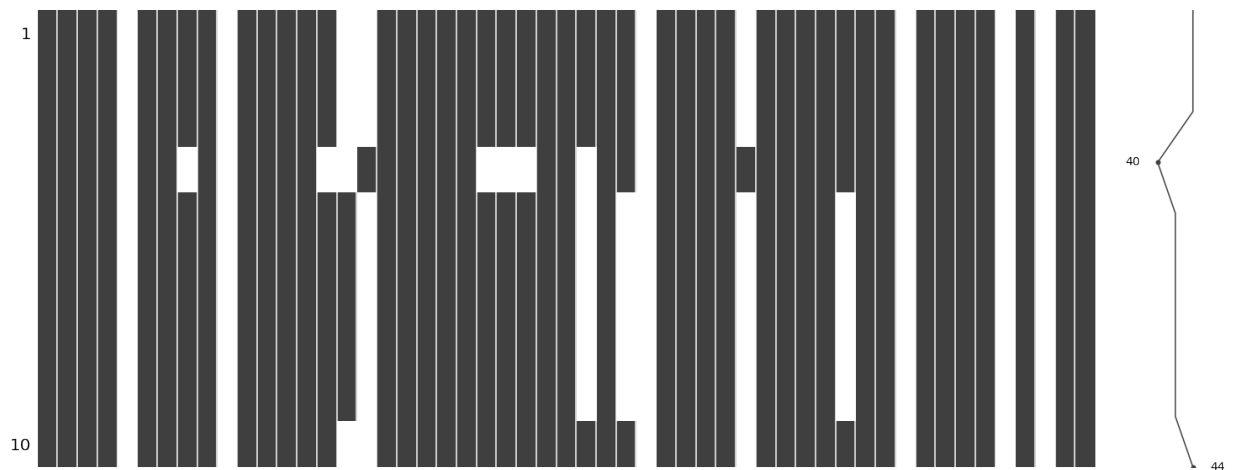
```
In [66]: tracks.drop(0, inplace=True)
```

```
In [67]: tracks.reset_index(inplace=True)
```

```
In [68]: tracks.drop(['index'], axis=1, inplace=True)
```

```
In [69]: ms.matrix(tracks.iloc[0: 10])
```

```
Out[69]: <AxesSubplot:>
```



```
In [70]: tracks.head()
```

```
Out[70]:
```

	track_id	album comments	album.1 date_created	album.2 date_released	album.3 engineer	album.4 favorites	album.5 id	album. informatio
0	2	0	2008-11-26 01:44:45	2009-01-05 00:00:00	NaN	4	1	
1	3	0	2008-11-26 01:44:45	2009-01-05 00:00:00	NaN	4	1	
2	5	0	2008-11-26 01:44:45	2009-01-05 00:00:00	NaN	4	1	
3	10	0	2008-11-26 01:45:08	2008-02-06 00:00:00	NaN	4	6	Na
4	20	0	2008-11-26 01:45:05	2009-01-06 00:00:00	NaN	2	4	"spiritu songs" froi Nick Cook

```
In [71]: tracks['track.7 genre_top'].value_counts()
```

```
Out[71]: Rock                14182
         Experimental        10608
         Electronic          9372
         Hip-Hop             3552
         Folk                2803
         Pop                 2332
         Instrumental        2079
         International       1389
         Classical          1230
         Jazz                571
         Old-Time / Historic  554
         Spoken              423
         Country             194
         Soul-RnB            175
         Blues               110
         Easy Listening        24
         Name: track.7 genre_top, dtype: int64
```

```
In [72]: track_title = pd.DataFrame(tracks['track.19 title'])
```

```
In [73]: track_title['track_id'] = tracks['track_id']
```

```
In [74]: track_title.head()
```

```
Out[74]:
```

	track.19 title	track_id
0	Food	2
1	Electric Ave	3
2	This World	5
3	Freeway	10
4	Spiritual Level	20

```
In [75]: track_title.tail()
```

```
Out[75]:
```

	track.19 title	track_id
106569	The Auger	155316
106570	Let's Skin Ruby	155317
106571	My House Smells Like Kim Deal/Pulp	155318
106572	The Man With Two Mouths	155319
106573	Another Trick Up My Sleeve (Instrumental)	155320

```
In [76]: track_title.shape
```

```
Out[76]: (106574, 2)
```

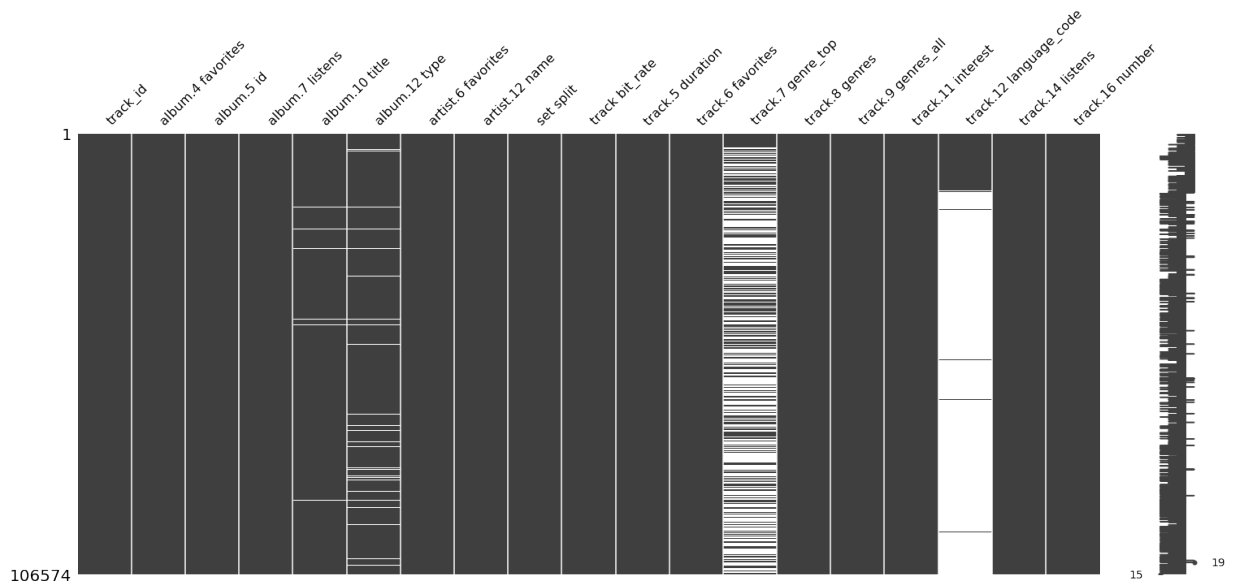
```
In [77]: tracks.drop(['album comments', 'album.1 date_created',
                    'album.2 date_released', 'album.11 tracks',
                    'album.9 tags', 'album.8 producer', 'album.3 engineer', '
                    'artist active_year_begin', 'artist.1 active_year_end', '
                    'artist.3 bio', 'artist.4 comments', 'artist.5 date_created
                    'artist.8 latitude', 'artist.9 location', 'artist.10 longit
                    'artist.13 related_projects', 'artist.14 tags', 'artist.15
                    'set.1 subset', 'track.1 comments', 'track.2 composer', '
                    'track.10 information', 'track.13 license', 'track.15 lyr
                    'track.19 title'], axis=1, inplace=True)
```

```
In [78]: tracks.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 106574 entries, 0 to 106573
Data columns (total 19 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   track_id                             106574 non-null object
1   album.4 favorites                    106574 non-null object
2   album.5 id                           106574 non-null object
3   album.7 listens                      106574 non-null object
4   album.10 title                       105549 non-null object
5   album.12 type                        100066 non-null object
6   artist.6 favorites                   106574 non-null object
7   artist.12 name                       106574 non-null object
8   set split                           106574 non-null object
9   track bit_rate                       106574 non-null object
10  track.5 duration                     106574 non-null object
11  track.6 favorites                    106574 non-null object
12  track.7 genre_top                    49598 non-null  object
13  track.8 genres                       106574 non-null object
14  track.9 genres_all                   106574 non-null object
15  track.11 interest                    106574 non-null object
16  track.12 language_code               15024 non-null  object
17  track.14 listens                     106574 non-null object
18  track.16 number                      106574 non-null object
dtypes: object(19)
memory usage: 15.4+ MB
```

```
In [79]: ms.matrix(tracks)
```

```
Out[79]: <AxesSubplot:>
```



```
In [80]: tracks['album.12 type'].value_counts()
```

```
Out[80]: Album                87549
Radio Program              6551
Live Performance          5035
Single Tracks              917
Contest                    14
Name: album.12 type, dtype: int64
```

```
In [81]: tracks['album.10 title'].value_counts()
```

```
Out[81]: microSong Entries                310
Sectioned v4.0                          200
Live at the 2014 Golden Festival         168
INTO INFINITY: an exploration of on and on and on... 151
Nectar Volume 6                        150
...
2999: The Age of Crickets                1
Live at The Parkside                    1
Bliss                                    1
Something in the Woods                  1
Live @ ISSUE (9/11/10)                  1
Name: album.10 title, Length: 14298, dtype: int64
```

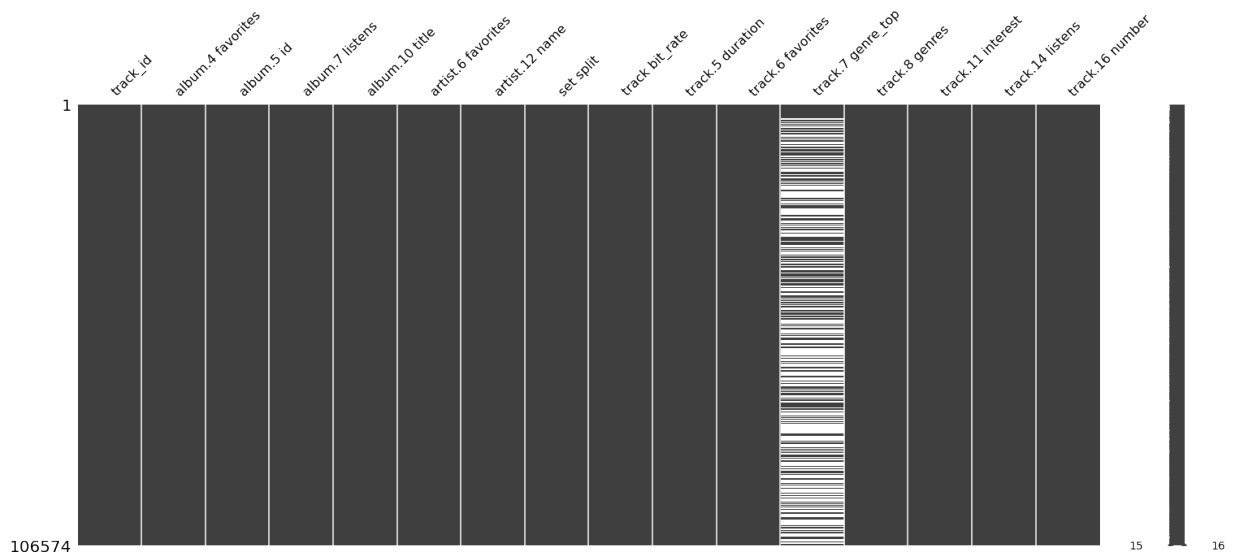
```
In [82]: tracks['album.10 title'].fillna(method='ffill', inplace=True)
```

```
In [83]: tracks.drop(['track.12 language_code', 'album.12 type'], axis=1, inplace=True)
```

```
In [84]: tracks.drop('track.9 genres_all', axis=1, inplace=True)
```

```
In [85]: ms.matrix(tracks)
```

```
Out[85]: <AxesSubplot:>
```



```
In [86]: tracks['track.8 genres'].unique()
```

```
Out[86]: array(['[21]', '[10]', '[76, 103]', ..., '[17, 63, 1235]',
               '[42, 107, 183]', '[10, 12, 169]'], dtype=object)
```

```
In [87]: genres.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 163 entries, 0 to 162
Data columns (total 5 columns):
#   Column      Non-Null Count  Dtype
---  -
0   genre_id    163 non-null   int64
1   #tracks     163 non-null   int64
2   parent      163 non-null   int64
3   title       163 non-null   object
4   top_level   163 non-null   int64
dtypes: int64(4), object(1)
memory usage: 6.5+ KB
```

```
In [88]: type(tracks['track.7 genre_top'].iloc[27])
```

```
Out[88]: str
```

```
In [89]: def getList(cd):
         return cd[1:-1].split(',')
```

```
In [90]: for i in range(0, 106574):
          if type(tracks['track.7 genre_top'][i]) == float:
              genre_list = getList(str(tracks['track.8 genres'][i]))
              count = len(genre_list)
              title = ""
              for j in range(0, count):
                  title = title + str(genres['title'][j]) + str('|')
              tracks['track.7 genre_top'][i] = title
```

```
In [91]: tracks['track.7 genre_top'].head()
```

```
Out[91]: 0          Hip-Hop
          1          Hip-Hop
          2          Hip-Hop
          3          Pop
          4  Avant-Garde|International|
Name: track.7 genre_top, dtype: object
```

```
In [92]: #Working with 'Genre' dataset
```

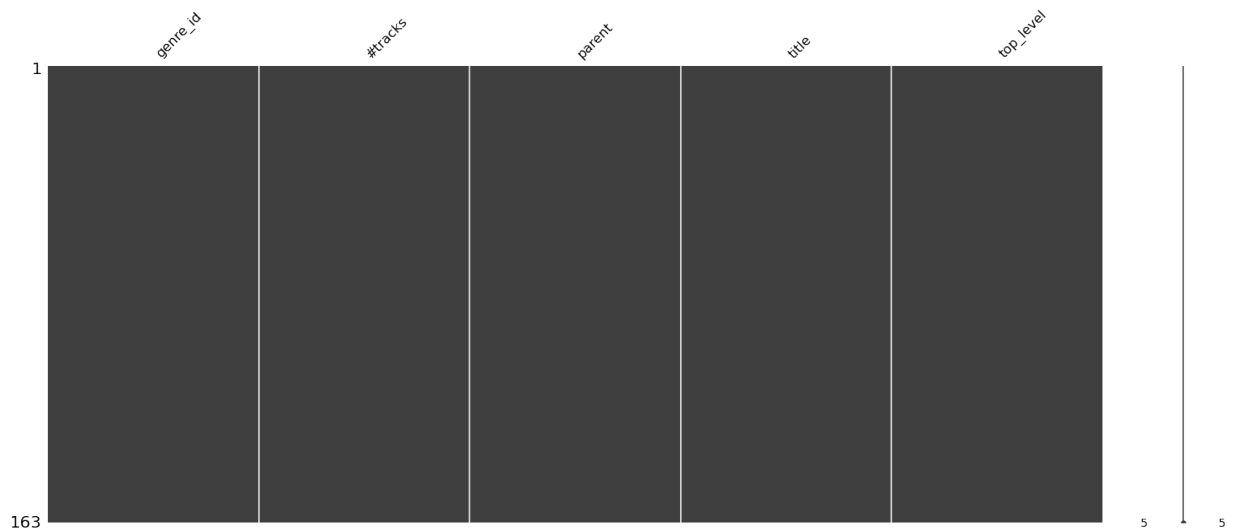
```
In [93]: genres.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 163 entries, 0 to 162
Data columns (total 5 columns):
#   Column      Non-Null Count  Dtype
---  -
0   genre_id    163 non-null   int64
1   #tracks     163 non-null   int64
2   parent      163 non-null   int64
3   title       163 non-null   object
4   top_level   163 non-null   int64
dtypes: int64(4), object(1)
memory usage: 6.5+ KB
```



```
In [94]: ms.matrix(genres)
```

```
Out[94]: <AxesSubplot:>
```



```
In [95]: genres.head()
```

```
Out[95]:
```

	genre_id	#tracks	parent	title	top_level
0	1	8693	38	Avant-Garde	38
1	2	5271	0	International	2
2	3	1752	0	Blues	3
3	4	4126	0	Jazz	4
4	5	4106	0	Classical	5

```
In [96]: #no feature engineering required
```

```
In [97]: #Combining all datasets into a single entity
          #Analysing Data
```

```
In [98]: echonest.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 13129 entries, 0 to 13128
Columns: 240 entries, track_id to 223
dtypes: float64(237), object(3)
memory usage: 24.0+ MB
```

In [99]: `tracks.info()`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 106574 entries, 0 to 106573
Data columns (total 16 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   track_id                             106574 non-null  object
1   album.4 favorites                     106574 non-null  object
2   album.5 id                           106574 non-null  object
3   album.7 listens                       106574 non-null  object
4   album.10 title                       106574 non-null  object
5   artist.6 favorites                   106574 non-null  object
6   artist.12 name                       106574 non-null  object
7   set split                            106574 non-null  object
8   track bit_rate                       106574 non-null  object
9   track.5 duration                     106574 non-null  object
10  track.6 favorites                     106574 non-null  object
11  track.7 genre_top                     106574 non-null  object
12  track.8 genres                       106574 non-null  object
13  track.11 interest                    106574 non-null  object
14  track.14 listens                     106574 non-null  object
15  track.16 number                       106574 non-null  object
dtypes: object(16)
memory usage: 13.0+ MB
```

In [100]: `tracks.head()`

Out[100]:

	track_id	album.4 favorites	album.5 id	album.7 listens	album.10 title	artist.6 favorites	artist.12 name	set split	track bit_rate	track duration
0	2	4	1	6073	AWOL - A Way Of Life	9	AWOL	training	256000	3:45
1	3	4	1	6073	AWOL - A Way Of Life	9	AWOL	training	256000	3:45
2	5	4	1	6073	AWOL - A Way Of Life	9	AWOL	training	256000	3:45
3	10	4	6	47632	Constant Hitmaker	74	Kurt Vile	training	192000	3:45
4	20	2	4	2710	Niris	10	Nicky Cook	training	256000	3:45

```
In [101]: echonest.head()
```

```
Out[101]:
```

	1	track_id	acousticness	danceability	energy	instrumentalness	liveness	speechiness	t
0	2	0.416675	0.675894	0.634476	0.010628	0.177647	0.159310	16	
1	3	0.374408	0.528643	0.817461	0.001851	0.105880	0.461818	12	
2	5	0.043567	0.745566	0.701470	0.000697	0.373143	0.124595	10	
3	10	0.951670	0.658179	0.924525	0.965427	0.115474	0.032985	11	
4	134	0.452217	0.513238	0.560410	0.019443	0.096567	0.525519	11	

```
In [102]: genres.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 163 entries, 0 to 162
Data columns (total 5 columns):
#   Column      Non-Null Count  Dtype
---  -
0   genre_id    163 non-null    int64
1   #tracks     163 non-null    int64
2   parent      163 non-null    int64
3   title       163 non-null    object
4   top_level   163 non-null    int64
dtypes: int64(4), object(1)
memory usage: 6.5+ KB
```

```
In [103]: features.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 106574 entries, 0 to 106573
Columns: 519 entries, feature track_id to zcr.6 std
dtypes: float64(518), int64(1)
memory usage: 422.0 MB
```

```
In [104]: #Feature Engineering
```

```
In [105]: features.columns = ['track_id'] + list(features.columns[1:])
```

```
In [106]: features.head()
```

```
Out[106]:
```

	track_id	chroma_cens kurtosis	chroma_cens.1 kurtosis	chroma_cens.2 kurtosis	chroma_cens.3 kurtosis	chroma_cens.4 kurtosis	ch
0	2	7.180653	5.230309	0.249321	1.347620	1.482478	
1	3	1.888963	0.760539	0.345297	2.295201	1.654031	
2	5	0.527563	-0.077654	-0.279610	0.685883	1.937570	
3	10	3.702245	-0.291193	2.196742	-0.234449	1.367364	
4	20	-0.193837	-0.198527	0.201546	0.258556	0.775204	

```
In [107]: type(echonest['track_id'].iloc[0])
```

```
Out[107]: str
```

```
In [108]: echonest['track_id'] = echonest['track_id'].astype('int')
tracks['track_id'] = tracks['track_id'].astype('int')
```

```
In [109]: features.sort_values(by='track_id', inplace=True)
tracks.sort_values(by='track_id', inplace=True)
echonest.sort_values(by='track_id', inplace=True)
```

```
In [110]: features.head()
```

```
Out[110]:
```

	track_id	chroma_cens kurtosis	chroma_cens.1 kurtosis	chroma_cens.2 kurtosis	chroma_cens.3 kurtosis	chroma_cens.4 kurtosis	ch
0	2	7.180653	5.230309	0.249321	1.347620	1.482478	
1	3	1.888963	0.760539	0.345297	2.295201	1.654031	
2	5	0.527563	-0.077654	-0.279610	0.685883	1.937570	
3	10	3.702245	-0.291193	2.196742	-0.234449	1.367364	
4	20	-0.193837	-0.198527	0.201546	0.258556	0.775204	

```
In [111]: tracks.head()
```

```
Out[111]:
```

	track_id	album.4 favorites	album.5 id	album.7 listens	album.10 title	artist.6 favorites	artist.12 name	set split	track bit_rate	trac dural
0	2	4	1	6073	AWOL - A Way Of Life	9	AWOL	training	256000	
1	3	4	1	6073	AWOL - A Way Of Life	9	AWOL	training	256000	
2	5	4	1	6073	AWOL - A Way Of Life	9	AWOL	training	256000	
3	10	4	6	47632	Constant Hitmaker	74	Kurt Vile	training	192000	
4	20	2	4	2710	Niris	10	Nicky Cook	training	256000	

```
In [112]: count = 0
for i in range(0, 106574):
    if features['track_id'][i] == tracks['track_id'][i]:
        count += 1
    else:
        print(features['track_id'][i], tracks['track_id'][i])
```

```
In [113]: final = pd.concat([features, tracks.drop('track_id', axis=1)], axis=1)
```

```
In [114]: final.shape
```

```
Out[114]: (106574, 534)
```

```
In [115]: final.head()
```

```
Out[115]:
```

	track_id	chroma_cens kurtosis	chroma_cens.1 kurtosis	chroma_cens.2 kurtosis	chroma_cens.3 kurtosis	chroma_cens.4 kurtosis	chr
0	2	7.180653	5.230309	0.249321	1.347620	1.482478	
1	3	1.888963	0.760539	0.345297	2.295201	1.654031	
2	5	0.527563	-0.077654	-0.279610	0.685883	1.937570	
3	10	3.702245	-0.291193	2.196742	-0.234449	1.367364	
4	20	-0.193837	-0.198527	0.201546	0.258556	0.775204	

```
In [116]: echonest.tail(3)
```

```
Out[116]:
```

	1	track_id	acousticness	danceability	energy	instrumentalness	liveness	speechiness
13126		124863	0.000124	0.609686	0.895136	0.846624	0.632903	0.05151
13127		124864	0.327576	0.574426	0.548327	0.452867	0.075928	0.03338
13128		124911	0.993606	0.499339	0.050622	0.945677	0.095965	0.06518

```
In [117]: echonest.drop(['artist_name', 'release'], axis=1, inplace=True)
```

```
In [118]: tracks.tail(3)
```

```
Out[118]:
```

	track_id	album.4 favorites	album.5 id	album.7 listens	album.10 title	artist.6 favorites	artist.12 name	set split	trac bit_rat
106571	155318	0	22940	1506	Live at Monty Hall, 2/17/2017	0	Spowder	training	32000
106572	155319	0	22940	1506	Live at Monty Hall, 2/17/2017	0	Spowder	training	32000
106573	155320	1	22906	7481	What I Tell Myself Vol. 2	12	Forget the Whale	validation	32000

```
In [119]: features.head(1)
```

```
Out[119]:
```

	track_id	chroma_cens kurtosis	chroma_cens.1 kurtosis	chroma_cens.2 kurtosis	chroma_cens.3 kurtosis	chroma_cens.4 kurtosis	ch
0	2	7.180653	5.230309	0.249321	1.34762	1.482478	

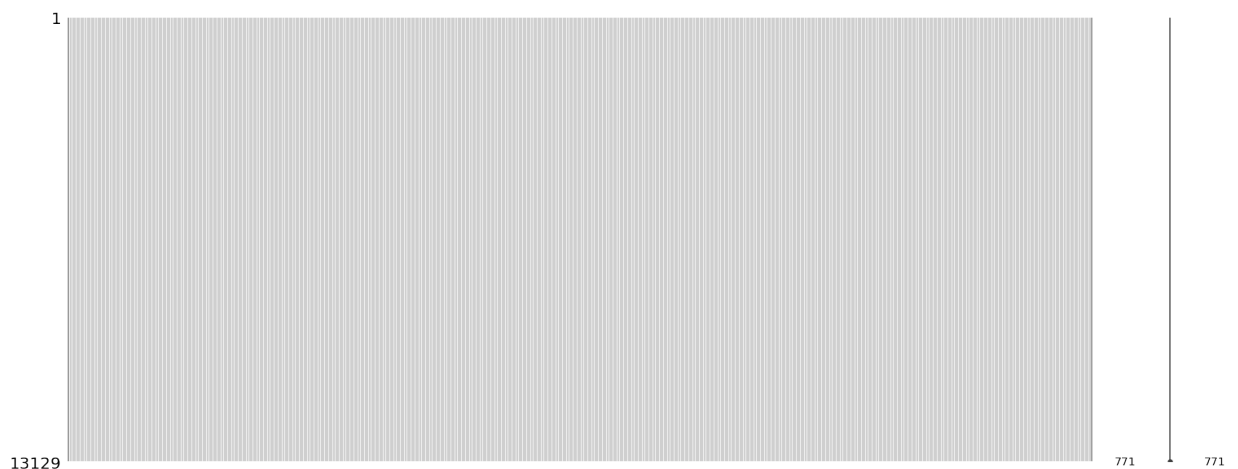
```
In [120]: final = echonest.merge(final, on='track_id')
```

```
In [121]: final.shape
```

```
Out[121]: (13129, 771)
```

```
In [122]: ms.matrix(final)
```

```
Out[122]: <AxesSubplot:>
```



```
In [123]: #Analysing Data
```

```
In [124]: final.head()
```

```
Out[124]:
```

	track_id	acousticness	danceability	energy	instrumentalness	liveness	speechiness	
0	2	0.416675	0.675894	0.634476	0.010628	0.177647	0.159310	16
1	3	0.374408	0.528643	0.817461	0.001851	0.105880	0.461818	12
2	5	0.043567	0.745566	0.701470	0.000697	0.373143	0.124595	10
3	10	0.951670	0.658179	0.924525	0.965427	0.115474	0.032985	10
4	134	0.452217	0.513238	0.560410	0.019443	0.096567	0.525519	10

```
In [125]: final.shape
```

```
Out[125]: (13129, 771)
```

```
In [126]: final.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Int64Index: 13129 entries, 0 to 13128
Columns: 771 entries, track_id to track.16 number
dtypes: float64(755), int64(1), object(15)
memory usage: 77.3+ MB
```

```
In [127]: final.drop('track.8 genres', axis=1, inplace=True)
```

```
In [128]: final.shape
```

```
Out[128]: (13129, 770)
```



```
In [129]: final.head()
```

```
Out[129]:
```

	track_id	acousticness	danceability	energy	instrumentalness	liveness	speechiness	
0	2	0.416675	0.675894	0.634476	0.010628	0.177647	0.159310	16
1	3	0.374408	0.528643	0.817461	0.001851	0.105880	0.461818	16
2	5	0.043567	0.745566	0.701470	0.000697	0.373143	0.124595	16
3	10	0.951670	0.658179	0.924525	0.965427	0.115474	0.032985	16
4	134	0.452217	0.513238	0.560410	0.019443	0.096567	0.525519	16

```
In [130]: final['track.7 genre_top'].value_counts()
```

```
Out[130]: Rock                                     3892
Electronic                                       2170
Avant-Garde|International|                     1902
Avant-Garde|International|Blues|               1507
Hip-Hop                                         910
Folk                                            874
Old-Time / Historic                           357
Pop                                             346
Avant-Garde|International|Blues|Jazz|          269
Classical                                     265
Jazz                                           241
International                                 133
Instrumental                                  84
Avant-Garde|International|Blues|Jazz|Classical| 82
Blues                                          66
Experimental                                  17
Avant-Garde|International|Blues|Jazz|Classical|Novelty| 14
Name: track.7 genre_top, dtype: int64
```

```
In [131]: #Feature Engineering
```

```
In [132]: final.head()
```

```
Out[132]:
```

	track_id	acousticness	danceability	energy	instrumentalness	liveness	speechiness	
0	2	0.416675	0.675894	0.634476	0.010628	0.177647	0.159310	16
1	3	0.374408	0.528643	0.817461	0.001851	0.105880	0.461818	16
2	5	0.043567	0.745566	0.701470	0.000697	0.373143	0.124595	16
3	10	0.951670	0.658179	0.924525	0.965427	0.115474	0.032985	16
4	134	0.452217	0.513238	0.560410	0.019443	0.096567	0.525519	16

```
In [133]: metadata = pd.DataFrame()
```

```
In [134]: metadata['track_id'] = final['track_id']
```

```
In [135]: metadata.shape
```

```
Out[135]: (13129, 1)
```

```
In [136]: track_title.shape
```

```
Out[136]: (106574, 2)
```

```
In [137]: track_title = track_title.set_index('track_id')
```

```
In [138]: track_title.head()
```

```
Out[138]:
```

track.19 title	
track_id	
2	Food
3	Electric Ave
5	This World
10	Freeway
20	Spiritual Level

```
In [139]: track_title.index = [int(i) for i in track_title.index]
```

```
In [140]: track_title.head()
```

Out[140]:

	track.19 title
2	Food
3	Electric Ave
5	This World
10	Freeway
20	Spiritual Level

```
In [141]: metadata.head()
```

Out[141]:

	track_id
0	2
1	3
2	5
3	10
4	134

```
In [142]: metadata['album_title'] = final['album.10 title']
```

```
In [143]: metadata['artist_name'] = final['artist.12 name']
```

```
In [144]: metadata['genre'] = final['track.7 genre_top']
```

```
In [145]: metadata = metadata.set_index('track_id')
```

```
In [146]: metadata.tail()
```

```
Out[146]:
```

	album_title	artist_name	genre
track_id			
124857	Do You Know The Word	BASIC	Avant-Garde International
124862	Do You Know The Word	BASIC	Avant-Garde International
124863	Do You Know The Word	BASIC	Avant-Garde International
124864	Do You Know The Word	BASIC	Avant-Garde International
124911	Suicide Beauty Girl	Yusuke Tsutsumi	Avant-Garde International

```
In [147]: metadata.head()
```

```
Out[147]:
```

	album_title	artist_name	genre
track_id			
2	AWOL - A Way Of Life	AWOL	Hip-Hop
3	AWOL - A Way Of Life	AWOL	Hip-Hop
5	AWOL - A Way Of Life	AWOL	Hip-Hop
10	Constant Hitmaker	Kurt Vile	Pop
134	AWOL - A Way Of Life	AWOL	Hip-Hop

```
In [148]: metadata['track_title'] = track_title.loc[metadata.index]['track.19 ti
```

```
In [149]: metadata.tail()
```

```
Out[149]:
```

	album_title	artist_name	genre	track_title
track_id				
124857	Do You Know The Word	BASIC	Avant-Garde International	I'm Wearing My Dancing Pants
124862	Do You Know The Word	BASIC	Avant-Garde International	Me Funky You Jane
124863	Do You Know The Word	BASIC	Avant-Garde International	Ain't Mad at Ya
124864	Do You Know The Word	BASIC	Avant-Garde International	This Is Not the End
124911	Suicide Beauty Girl	Yusuke Tsutsumi	Avant-Garde International	Heaven

```
In [150]: metadata.head()
```

```
Out[150]:
```

	album_title	artist_name	genre	track_title
track_id				
2	AWOL - A Way Of Life	AWOL	Hip-Hop	Food
3	AWOL - A Way Of Life	AWOL	Hip-Hop	Electric Ave
5	AWOL - A Way Of Life	AWOL	Hip-Hop	This World
10	Constant Hitmaker	Kurt Vile	Pop	Freeway
134	AWOL - A Way Of Life	AWOL	Hip-Hop	Street Music

```
In [151]: len(metadata[metadata['genre'].isnull()])
```

```
Out[151]: 0
```

```
In [152]: final.drop('album.10 title', axis=1, inplace=True)
```

```
In [153]: final.head()
```

```
Out[153]:
```

	track_id	acousticness	danceability	energy	instrumentalness	liveness	speechiness	1
0	2	0.416675	0.675894	0.634476	0.010628	0.177647	0.159310	16
1	3	0.374408	0.528643	0.817461	0.001851	0.105880	0.461818	12
2	5	0.043567	0.745566	0.701470	0.000697	0.373143	0.124595	10
3	10	0.951670	0.658179	0.924525	0.965427	0.115474	0.032985	1
4	134	0.452217	0.513238	0.560410	0.019443	0.096567	0.525519	1

```
In [154]: final.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Int64Index: 13129 entries, 0 to 13128
Columns: 769 entries, track_id to track.16 number
dtypes: float64(755), int64(1), object(13)
memory usage: 77.1+ MB
```

```
In [155]: final.drop('artist.12 name', axis=1, inplace=True)
```

```
In [156]: final.head()
```

```
Out[156]:
```

	track_id	acousticness	danceability	energy	instrumentalness	liveness	speechiness	1
0	2	0.416675	0.675894	0.634476	0.010628	0.177647	0.159310	16
1	3	0.374408	0.528643	0.817461	0.001851	0.105880	0.461818	12
2	5	0.043567	0.745566	0.701470	0.000697	0.373143	0.124595	16
3	10	0.951670	0.658179	0.924525	0.965427	0.115474	0.032985	17
4	134	0.452217	0.513238	0.560410	0.019443	0.096567	0.525519	17

```
In [157]: k = final # Restore point # Removed Label Encoding
```

```
In [158]: final.drop('set split', axis=1, inplace=True)
```

```
In [159]: final.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Int64Index: 13129 entries, 0 to 13128
Columns: 767 entries, track_id to track.16 number
dtypes: float64(755), int64(1), object(11)
memory usage: 76.9+ MB
```

```
In [160]: final.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Int64Index: 13129 entries, 0 to 13128
Columns: 767 entries, track_id to track.16 number
dtypes: float64(755), int64(1), object(11)
memory usage: 76.9+ MB
```

```
In [161]: genres['title'].count()
```

```
Out[161]: 163
```

```
In [162]: genre_dummy = pd.DataFrame(data= np.zeros((13129, 163)), columns= list
```

```
In [163]: genre_dummy.head()
```

Out[163]:

	Avant-Garde	International	Blues	Jazz	Classical	Novelty	Comedy	Old-Time / Historic	Country	Pop	Disco
0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

```
In [164]: genre_list = pd.Series(data= genre_dummy.columns)
```

```
In [167]: genre_dummy.columns= genre_list
```

```
In [168]: genre_list = list(genre_list)
```

```
In [169]: final
```

Out[169]:

	track_id	acousticness	danceability	energy	instrumentalness	liveness	speechiness
0	2	0.416675	0.675894	0.634476	0.010628	0.177647	0.159310
1	3	0.374408	0.528643	0.817461	0.001851	0.105880	0.461810
2	5	0.043567	0.745566	0.701470	0.000697	0.373143	0.124590
3	10	0.951670	0.658179	0.924525	0.965427	0.115474	0.032980
4	134	0.452217	0.513238	0.560410	0.019443	0.096567	0.525510
...	...	...	...	...	...	...	...
13124	124857	0.007592	0.790364	0.719288	0.853114	0.720715	0.082550
13125	124862	0.041498	0.843077	0.536496	0.865151	0.547949	0.074000
13126	124863	0.000124	0.609686	0.895136	0.846624	0.632903	0.051510
13127	124864	0.327576	0.574426	0.548327	0.452867	0.075928	0.033380
13128	124911	0.993606	0.499339	0.050622	0.945677	0.095965	0.065180

13129 rows × 767 columns

```
In [170]: for i in range(0, 13129):
            if '|' in final['track.7 genre_top'][i]:
                divided_list = str(final['track.7 genre_top'][i]).split('|')
                count = len(divided_list)
                for j in range(0, count):
                    if divided_list[j] in genre_list:
                        location = genre_list.index(divided_list[j])
                        genre_dummy.iloc[i, location] = 1
            else:
                location = genre_list.index(final['track.7 genre_top'][i])
                genre_dummy.iloc[i, location] = 1
```

```
In [171]: genre_list.index(final['track.7 genre_top'][0])
```

```
Out[171]: 20
```

```
In [172]: final.drop(['track.7 genre_top'], axis= 1, inplace= True)
```

```
In [173]: final = pd.concat([final, genre_dummy], axis= 1)
```

```
In [174]: final.head()
```

```
Out[174]:
```

	track_id	acousticness	danceability	energy	instrumentalness	liveness	speechiness	1
0	2	0.416675	0.675894	0.634476	0.010628	0.177647	0.159310	16
1	3	0.374408	0.528643	0.817461	0.001851	0.105880	0.461818	16
2	5	0.043567	0.745566	0.701470	0.000697	0.373143	0.124595	16
3	10	0.951670	0.658179	0.924525	0.965427	0.115474	0.032985	16
4	134	0.452217	0.513238	0.560410	0.019443	0.096567	0.525519	16

```
In [ ]: import os

            if not os.path.isdir(os.path.join('datasets', 'final')):
                os.makedirs(os.path.join('datasets', 'final'))

            metadata.to_csv('datasets/final/metadata.csv')
            final.to_csv('datasets/final/final.csv')
```